

ESILV DATA & AI · MASTER THESIS

Anomaly detection for data quality, measured inside the ETL pipeline

Rules, classical and neural detectors run through one pipeline and compared per anomaly type. Living progress deck, 2026-11-12.

Illustration, not data: a generated still life. One pad of 312 stands proud; the low light finds it.

17 DETECTORS 6 ANOMALY TYPES 152 DATASETS

Theo Bigand · github.com/theobigand/etl-anomaly-benchmark

THE QUESTION

Does machine learning in an ETL pipeline beat plain rules, and for which anomalies?

Rules are the industry default: auditable, cheap, and blind to what the author did not anticipate. Recent work puts ML detectors inside the flow, but on one algorithm and one dataset. This thesis builds the missing head-to-head benchmark and breaks the result down by anomaly type.

WHAT GENERIC BENCHMARKS LEAVE OUT

Pipeline. Algorithms are scored outside any pipeline; integration cost is not measured.

Rules. The declarative baseline that dominates production is absent.

Types. One aggregate score per dataset hides which family finds which defect.

Threshold. One flag threshold, not the precision-recall trade-off.

Benchmarks rank detectors. Few report their decisions

Work	Time order	At a threshold
ADBench, 30 algorithms, 57 sets	not found	not found (AUC)
Agyemang 2024, one simulated set	none	precision, recall, F1
AnoShift, Kyoto 2006 to 2015	IID, NEAR, FAR	not found (AUC)
FedAD-Bench, federated deep	not found	F1-optimal threshold
Sun et al. 2024, online thresholds	shift in a stream	error-rate guarantees

In the texts I read, ADBench and AnoShift report AUC only, and none of these works breaks results down by data-quality fault.

The scan is not systematic: this is a gap I did not find filled, not a proof that nothing exists. References were checked against a primary source or the DOI registry.

"Not found" means absent from the text I read, not proven absent. This thesis: six fault types, a controlled generator and real tables, decisions and pipeline cost.

RESEARCH QUESTIONS

Four questions, one study each, hypotheses written first

RQ1	**Rules, classical and neural detectors in one pipeline** examples, 100 labelled problems	historical ranking, train-only replay pending
RQ2	**Which anomaly type each family finds** synthetic matrix, schemas, 42 UCI tables	narrow competence; values do not transfer
RQ3	**Operating point and time** contamination, label-free threshold, chronological split	a threshold fixed on the past fails when prevalence moves
RQ4	**Cost in the pipeline** cost study, ETL run on 152 sets	neural 10 to 100 times column-wise

Every study has a protocol and hypotheses committed before its runs. A hypothesis that fails is reported as failed: three of the six on the whole pool do.

`paper, Introduction and Methods; pre-registered in verification/*.md`

THE ORIGINAL CONTRIBUTION

A controlled dataset that labels six anomaly types separately

12,360 records generated with Faker and NumPy, six types injected at 3% each, every row tagged with its category. Recall and precision are reported per type, which generic benchmarks that publish one aggregate score do not provide.

The magnitudes are one severity multiplier, so the choice is swept rather than fixed. Glyphs: a pad that is missing, doubled, raised, shifted, sloping or cracked.



missing value

one or two feature cells set to null



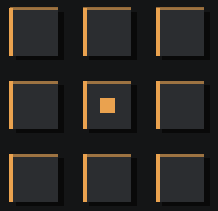
exact duplicate

a clean row copied, same business attributes



univariate outlier

one numeric field pushed far out of range



multivariate outlier

every field in range, the combination not



temporal drift

recent signups whose income and balance have shifted up



invalid format

malformed e-mail, out-of-domain country code, impossible age

`make synthetic` · `make severity` · generator is deterministic, dataset not committed

One interface, one measured path

extract
read the table

transform
exact dedup, clean

detect
fit + predict + score

load
write the files

latency is measured here, in the real in-pipeline time

The demo and every benchmark call the same `detect()` stage. A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and AP computable; `detect(fit_df=...)` fits on the train slice and scores the test set.

Exact deduplication belongs to the ordinary transform. Typed injected duplicates remain in their separate evaluation with transform_dedup.

```
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:  # bool/row
        ...
    def score(self, df) -> Series:    # graded
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector): ...

# registry: one line per detector
"isolation_forest": lambda: IsolationForestDetector(),
```

METRIC INTERPRETATION

Average Precision measures the ranking

$$AP = \sum \Delta R \cdot P$$

precision, weighted by each recall increment, over distinct score cutoffs from high to low; tied scores enter together. AP uses continuous anomaly scores. Precision, recall, F1 and the flagged share use a selected threshold.

Non-interpolated

AP differs from trapezoidal precision-recall area.

Prevalence matters

Prevalence is the no-skill reference. Constant scores give AP equal to prevalence.

Decisions apart from scores

Changing only the flag threshold leaves AP and ROC-AUC unchanged; neither fixes a useful operating point.

Historical naming: stored pr_auc fields and remaining AP labels in archived figures and the paper denote this same AP quantity.

Rules and classical models, one contract

Detector	Family	Principle	Main limit
Z-score	rule	flag |z| > 3 on any column	mean and std inflated by outliers
IQR	rule	Tukey fences, k = 1.5	fires often in high dimension
Great Expectations	rule	null checks, fitted 0.1% / 99.9% bounds	generic suite, no domain rules
Isolation Forest	joint	isolation depth under random splits	axis-aligned splits, weak on nulls
LOF	joint	local density against 20 neighbours	cost grows with the scored set
One-Class SVM	joint	RBF boundary around the bulk	quadratic training
ECOD	column-wise	empirical tail probabilities, summed	no explicit cross-feature dependence
COPOD	column-wise	max of skew term and mean log tails	same
HBOS	column-wise	histogram per column, inverse bin height	independent columns, coarse bins
KNN	joint	distance to the 5th neighbour	quadratic, skipped above 50k rows
PCA	joint	reconstruction error, 95% of variance	misses a shift inside the kept space
drift	time-aware	recent-window mean shift	per row, so diluted drift is invisible

PyOD or the recorded protocol adaptations; flag threshold from training prevalence. Numeric columns only, while Great Expectations separately checks nulls on all columns: an explicit information asymmetry. The time-aware drift detector is opt-in and synthetic only.

`src/etl_benchmark/quality/ml.py, rules.py, gx.py · ECOD/COPOD use training ranks and frozen skew signs`

FITTED RULES

Column bounds do not fix row alert rates

Z-score

Maximum |x − training mean| / sample standard deviation. Flag above 3; constant columns contribute zero.

IQR

Training fences Q1 − 1.5 IQR and Q3 + 1.5 IQR. Flag strictly outside; zero-IQR columns contribute zero.

Great Expectations

Not-null on all columns; inclusive training quantiles 0.1% and 99.9%. Score: fraction of active checks violated.

GAUSSIAN REFERENCE ONLY

Per-column tails: **0.27%** beyond 3σ, **0.70%** beyond the IQR fences (about 2.7σ). Fitted bounds and several columns change row-level rates.

SCOPE

All statistics come from training data. The evaluated Great Expectations suite is a generic statistical baseline: no domain-specific category, format or cross-field rules.

Neural detectors behind the same contract

Detector	Principle	Fixed or noted
Autoencoder	rebuild standardized rows through a bottleneck	MLP from scikit-learn
VAE	Gaussian code, rebuild, reconstruction error	scored with the code mean; PyOD draws one
Deep SVDD	map normal rows near a centre, distance	weakest on most types
DIF	isolation forests on random neural features	training representation scaler reused
AE1SVM	autoencoder + random-Fourier one-class SVM	singleton padding changes training weight
LUNAR	network learns the score from neighbour distances	best on drift, many false alarms

PyOD 3.6.7 with recorded adaptations, CPU, seeded, fitted on at most 20,000 rows.

Frozen representation scaling and latent-mean scoring change the evaluated algorithms. They do not establish equivalence to upstream defaults.

The LSTM autoencoder reads windows of ordered rows and stays outside the registry.

EVALUATED VARIANTS

Stable row scores require explicit adaptations

A fitted ETL detector must score a row without consulting other evaluation rows. Contract tests check the 17 row-level detectors; the time-window monitor has a different contract.

EVIDENCE BOUNDARY

Bounded tests support batch independence, not universal equivalence. Separate ablations remain open; historical results keep their actual adapted configurations.

ECOD and COPOD

PyOD 3.6.7 recomputes tails and skew with the evaluated batch. Here ranks and skew signs are frozen from training.

Copula motivation versus score

COPOD adds marginal contributions: it is not a fitted general joint copula. ECOD takes the max of both log tails and the skew term.

Neural scoring

VAE decodes the latent mean; DIF reuses a training scaler; AE1SVM padding reweights its last row. These are algorithm changes.

Historical pool preparation needs revalidation

Historical path

Column filtering, category vocabulary and conflicting-label removal happen **before** the split. Training-only medians and scaling do not isolate those earlier decisions.

Constructed negative controls: changing only the evaluation categories changes the encoded width from 5 to 3; an evaluation duplicate's label can remove a reference row. The effect on AP is not yet quantified.

pool-train-only-v1

Consumes source-bound frozen positions. Fitted schema, imputation and scaling use training rows only. Cross-partition duplicates are rejected; evaluation rows are retained. Targets are excluded from preparation.

Bounded checks enforce full line and branch coverage for the two new modules and detect three leakage defects.

STILL OUTSTANDING

Historical memberships were recomputed (not originally archived) and replay-validated: the Z-score reproduces all 100 archived per-seed results. **Replaying the studies on the train-only path remains outstanding.** Old protocols and results are preserved, and no corrected pool-performance claim is available.

PART 2 OF 4

What the detectors actually find

Three example datasets, then six anomaly types on a controlled generator and on real tables.

Three example datasets bracket the method

Dataset	Rows	Feat.	Anom.	Role
Credit Card Fraud	283,726	30	0.17%	rare, but a mixed-defect label
Cardio	1,831	21	9.6%	rare and outlying, the fitting regime
Water Quality	3,276	9	61%	target class common, wrong regime
Synthetic per-type	12,360	9	17.5%	controlled, 6 labelled types

0.01→0.20

Isolation Forest fraud precision, from the contamination change alone

Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std. The ML contamination is read from the train-set prevalence, not fixed at 10%.

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

The magnitudes are a swept parameter, not a fixed choice

Type	Injection
missing value	1 to 2 of {income, balance, avg amount, tenure, n txn, email, country} set to null
duplicate	exact copy of a sampled clean row, id included (appended, not replaced)
univariate	one numeric column × U(15, 45) × severity
multivariate	marginals at the (1 − 0.03 / severity) percentile of the clean data
temporal drift	signup in the last 80 days; income × (1 + 0.55 · severity), similar for balance and amount
invalid format	email malformed; age impossible; country out of domain

Severity 1.0 is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale. Five types replace clean rows; duplicates are appended. Sweeping the multiplier turns "we picked the magnitudes" into a curve, and the per-type ranking does not change (severity slide).

Aggregate scores do not establish equivalence.

Per-type results show different competencies.

F1 spans 0.22 to 0.59 over the seventeen detectors on the synthetic set, and the per-type competences barely overlap.

The operating point decides the outcome

Detector	Prec	Rec	F1	ROC	PR	Flag %	Time
Z-score	0.011	0.870	0.021	0.926	0.078	13.4	0.03 s
IQR	0.003	**0.966**	0.007	0.911	0.018	48.6	0.12 s
Great Expectations	0.037	0.676	0.070	0.830	**0.182**	3.1	1.37 s
Isolation Forest	**0.202**	0.206	**0.203**	**0.934**	0.120	0.2	1.78 s

best of column

The statistical rules recover most frauds at precision below 0.04. Isolation Forest changes once its contamination matches the prevalence: it flags 0.17% of rows and reaches the first usable precision, at recall 0.21 and with real seed variance (F1 = 0.20 ± 0.07). Great Expectations leads AP.

0.20

Isolation Forest precision (0.01 at contamination 0.10)

0.182

Great Expectations AP, the best on this dataset

Mean over 5 seeds, 15% held-out test split. LOF and One-Class SVM do not scale to 284k rows here.

`experiments/run_benchmark.py · make benchmark · the metric that matters here: AP`

GENUINELY IMBALANCED REAL DATA

When anomalies are rare and outlying, the learned detectors work

Detector	Prec	Rec	F1	ROC	PR	Flag %
Z-score	0.49	0.82	0.61	0.93	0.61	16.6
IQR	0.21	**0.90**	0.34	0.78	0.30	43.4
Great Expectations	0.51	0.21	0.29	0.59	0.19	4.6
Isolation Forest	**0.64**	0.62	**0.62**	**0.95**	**0.70**	9.9
LOF	0.25	0.27	0.26	0.62	0.24	10.7
One-Class SVM	0.56	0.61	0.58	0.94	0.63	10.7

best of column

0.62

Isolation Forest F1

0.70

Isolation Forest AP

Isolation Forest reaches F1 0.62 and AP 0.70, with ECOD, COPOD and DIF close behind (AP 0.69, 0.69, 0.67). Z-score also does well: the 21 features are standardized and anomalies stand out on several at once.

Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split. Wide seed spread on this small set (ML F1 std up to 0.17).

`make benchmark-cardio` · the regime the detectors are built for

Aggregate scores hide different trade-offs

Detector	Prec	Rec	F1	ROC	PR
RULES AND CLASSICAL					
Z-score	**1.000**	0.196	0.328	0.742	0.544
IQR	0.584	0.396	0.472	0.675	0.452
Isolation Forest	0.495	0.499	0.497	0.735	0.485
One-Class SVM	0.478	0.485	0.481	0.722	0.520
ECOD	**0.341**	0.341	0.341	0.675	0.283
HBOS	1.000	0.204	0.339	0.602	0.343
KNN	0.466	0.469	0.467	0.745	0.520
NEURAL					
VAE	0.502	0.507	0.504	0.742	0.548
DIF	0.589	0.582	**0.585**	**0.772**	**0.648**
LUNAR	0.345	**0.668**	0.410	0.728	0.495

Z-score and HBOS reach a perfect precision by firing only on extreme rows (recall 0.20). DIF has the best F1 and AP, the VAE matches Isolation Forest, LUNAR has the best recall with a low precision.

0.22 to 0.59

F1 range across the seventeen detectors

Mean over 5 seeds, test split. 12,360 rows, 17.5% anomalous. All 17 detectors are in the paper, table tab:synthetic. The per-type view shows the opposite.

best of column

verification/evidence/full_registry · paper, tab:synthetic

PER TYPE, THE CONTRIBUTION

Each method has a narrow competence

Z-score owns univariate outliers (0.93). **Great Expectations** owns missing values (1.00). **Deduplication** owns exact duplicates.

Isolation Forest (0.89) and the One-Class SVM (0.90) are the only first-six methods with real recall on multivariate outliers; every rule stays at 0.04 or below.

Broad coverage needs several families, not the single best one.

	missing	duplicate	univariate	multivariate	drift	invalid
Z-score	0.00	0.00	0.93	0.00	0.00	0.24
IQR	0.05	0.04	0.99	0.04	0.25	1.00
GE	1.00	0.00	0.21	0.00	0.00	0.00
IF	0.09	0.11	0.99	0.89	0.33	0.60
LOF	0.28	0.16	0.70	0.21	0.50	0.67
OCSVM	0.07	0.10	1.00	0.90	0.29	0.54
ECOD	0.10	0.13	0.68	0.52	0.26	0.36
COPOD	0.09	0.13	0.89	0.51	0.34	0.30
HBOS	0.00	0.00	0.90	0.00	0.00	0.32
KNN	0.09	0.10	1.00	0.39	0.41	0.81
PCA	0.10	0.12	1.00	0.48	0.23	0.58
AE	0.12	0.10	0.81	0.83	0.29	0.53
VAE	0.07	0.11	0.99	0.92	0.30	0.66
DSVDD	0.19	0.18	0.33	0.13	0.16	0.32
DIF	0.09	0.07	1.00	1.00	0.33	1.00
AE1SVM	0.09	0.10	0.99	0.78	0.27	0.61
LUNAR	0.44	0.39	1.00	0.63	0.65	0.90

Amber intensity is mean recall, 0 to 1.

paper, Appendix C, tab:m-customer · synthetic customer table, mean recall over 5 seeds · duplicates need the transform stage

RECALL AND PRECISION, PER TYPE

Detection rate alone can be gamed

A wide net

0.89 recall

~0.22 precision

Isolation Forest and the One-Class SVM on multivariate outliers: they cast a wide net that also catches clean rows.

A narrow one

1.00 precision

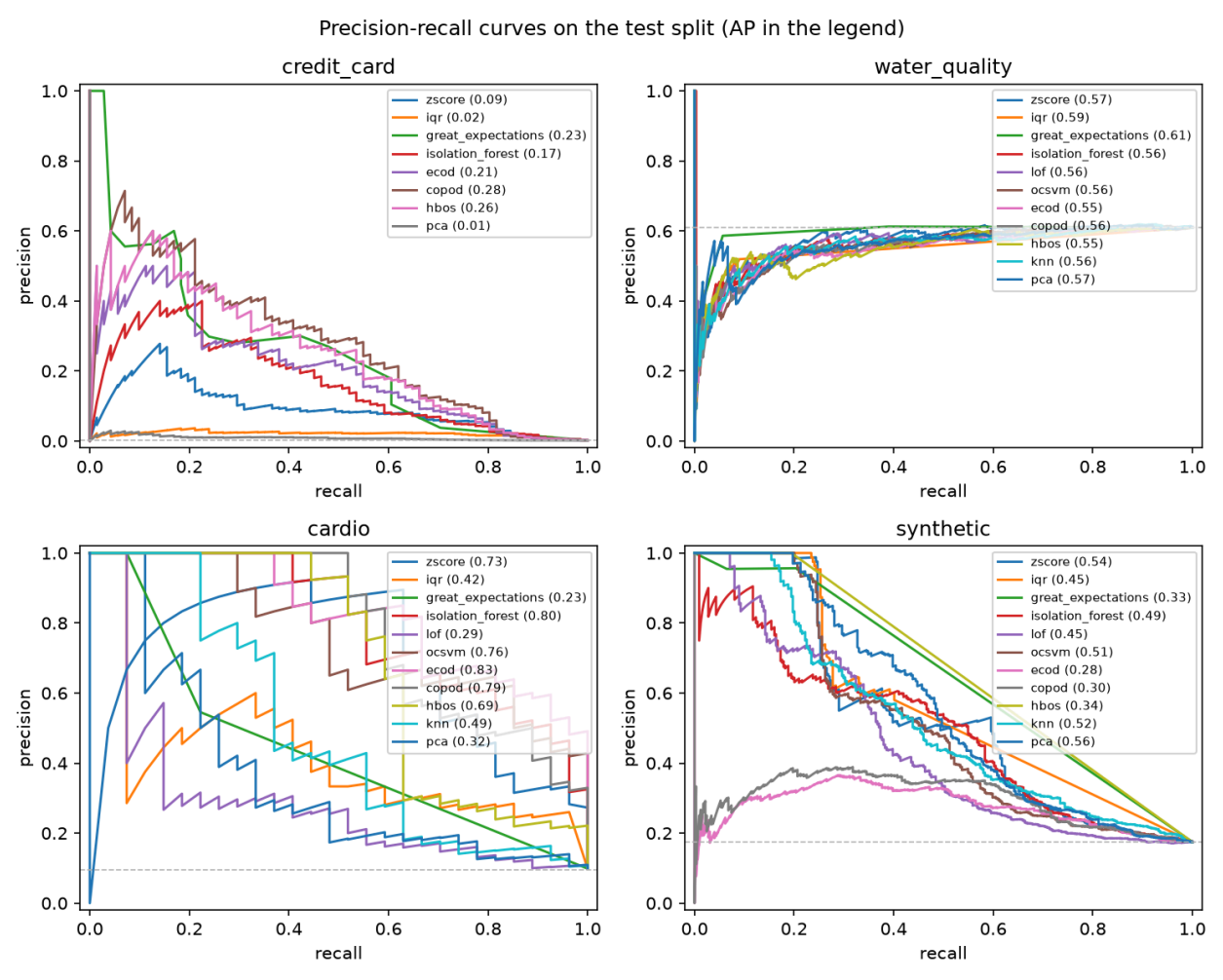
0.20 recall

Z-score fires only when it is certain: on the whole synthetic set its precision is 1.00 and its recall 0.196.

Clean rows are the only negatives. Both views are needed: neither number alone ranks the detectors.

paper, per-type results and tab:synthetic · synthetic customer table, 5 seeds

One threshold hides the whole trade-off

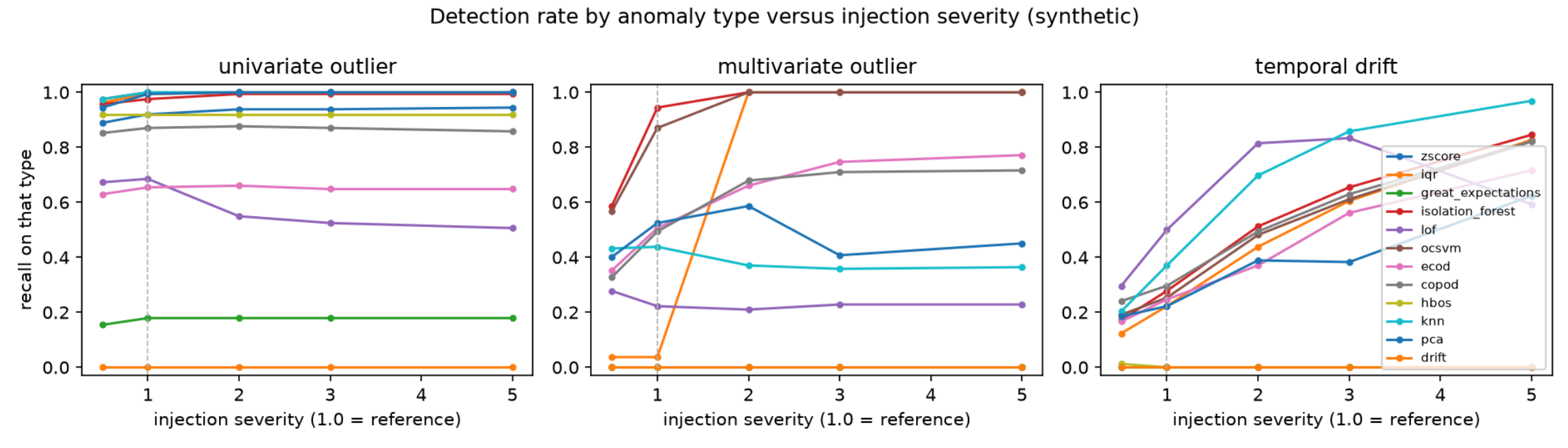


On **Cardio** the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On **Water Quality** every curve hugs the 0.61 base-rate line. On **Credit Card Fraud** two methods dominate but neither clears precision 0.4 except at negligible recall.

0.37 → 0.008

Isolation Forest fraud precision as recall goes 0.23 to 0.92

The per-type ranking is stable as the magnitudes scale



Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold. Temporal drift climbs roughly linearly for the distributional detectors: the reference drift is below every method's floor, not a property of the methods. The same families lead the same columns at every severity.

≤0.3→0.85

temporal-drift recall from severity 1 to 3

PART 3 OF 4

From three examples to a pool of 152

Real tables, pinned and checked; what holds, what fails, and what changes with time.

A pool of 152 open datasets, pinned and checked

ADBENCH

47 tabular sets in the commit

36 kept: terms read and open

11 excluded: no open licence or terms forbid use (celeba, mammography, vowels ...)

UCI REPOSITORY

689 catalogue entries

133 pass metadata and licence filters

116 kept after curation

17 removed with a reason (version duplicates, rule-labelled, boards)

36 + 116

= **152** pinned datasets: ADBench kept plus UCI curated

116

UCI tables as clean carriers for typed injection

36 + 64

= **100** labelled problems; rule fixed before any detector: the rarest class is the anomaly

One normalisation policy for all sets, applied in order. 19 sources appear in both lists with a different derivation. Raw data stays out of the repository; checksums, licences and reports are archived. A download check of every set runs manually and monthly.

Historical pool rankings need train-only revalidation

Detector	Problems	AP median [95%]	ROC-AUC	Rank
ECOD	100	0.240 [0.191, 0.305]	0.656	7.0
One-Class SVM	84	0.240 [0.177, 0.307]	0.670	8.0
HBOS	100	0.237 [0.176, 0.311]	0.663	8.0
DIF	100	0.226 [0.173, 0.272]	0.639	8.0
VAE	100	0.216 [0.163, 0.279]	0.676	7.0
COPOD	100	0.216 [0.162, 0.314]	0.691	7.0
Isolation Forest	100	0.215 [0.158, 0.293]	0.677	7.0
AE1SVM	100	0.210 [0.172, 0.253]	0.635	8.0
Autoencoder	100	0.205 [0.150, 0.240]	0.631	8.0
Z-score	100	0.203 [0.164, 0.279]	0.626	9.0
KNN	92	0.203 [0.172, 0.284]	0.658	8.0
LUNAR	100	0.183 [0.153, 0.225]	0.615	10.0
Deep SVDD	100	0.181 [0.147, 0.226]	0.619	10.0
PCA	100	0.178 [0.152, 0.219]	0.581	10.0
LOF	92	0.165 [0.153, 0.237]	0.588	10.0
IQR	100	0.144 [0.117, 0.191]	0.556	12.0
Great Expectations	100	0.115 [0.105, 0.123]	0.508	16.0

Three rules, eight classical and six neural detectors, five seeds, on 36 ADBench and 64 UCI problems. The table is retained as recorded.

Schema and row filtering precede the split, so these descriptive rankings still need train-only replay.

The 100 problems come from 80 sources; a source-clustered re-analysis changes no sign (only Great Expectations is clearly below Isolation Forest). Label-rule sensitivity and unrecorded PyOD convergence diagnostics remain open.

17

detectors; no neural exception

13

problems won by DIF, the most

Median over problems, sorted by AP. LOF and KNN are skipped above 50,000 training rows, One-Class SVM above 20,000; neural models fit on at most 20,000 rows.

Two regimes the three examples do not show

AP	Credit Card	Cardio	Annthyroid	Mushroom
Z-score	0.10	0.61	0.18	0.44
IQR	0.02	0.30	**0.61**	0.10
Isolation Forest	0.13	**0.70**	0.29	0.63
ECOD	0.21	0.69	0.28	0.41
HBOS	**0.26**	0.58	0.24	0.50
Autoencoder	0.11	0.35	0.14	0.68
DIF	0.27	0.67	0.23	0.45
LUNAR	0.15	0.33	0.20	**0.96**

best of column

Chosen after the results, as illustrations, not as a test. Mean over 5 seeds, as run in the pool. All 17 detectors are in the paper.

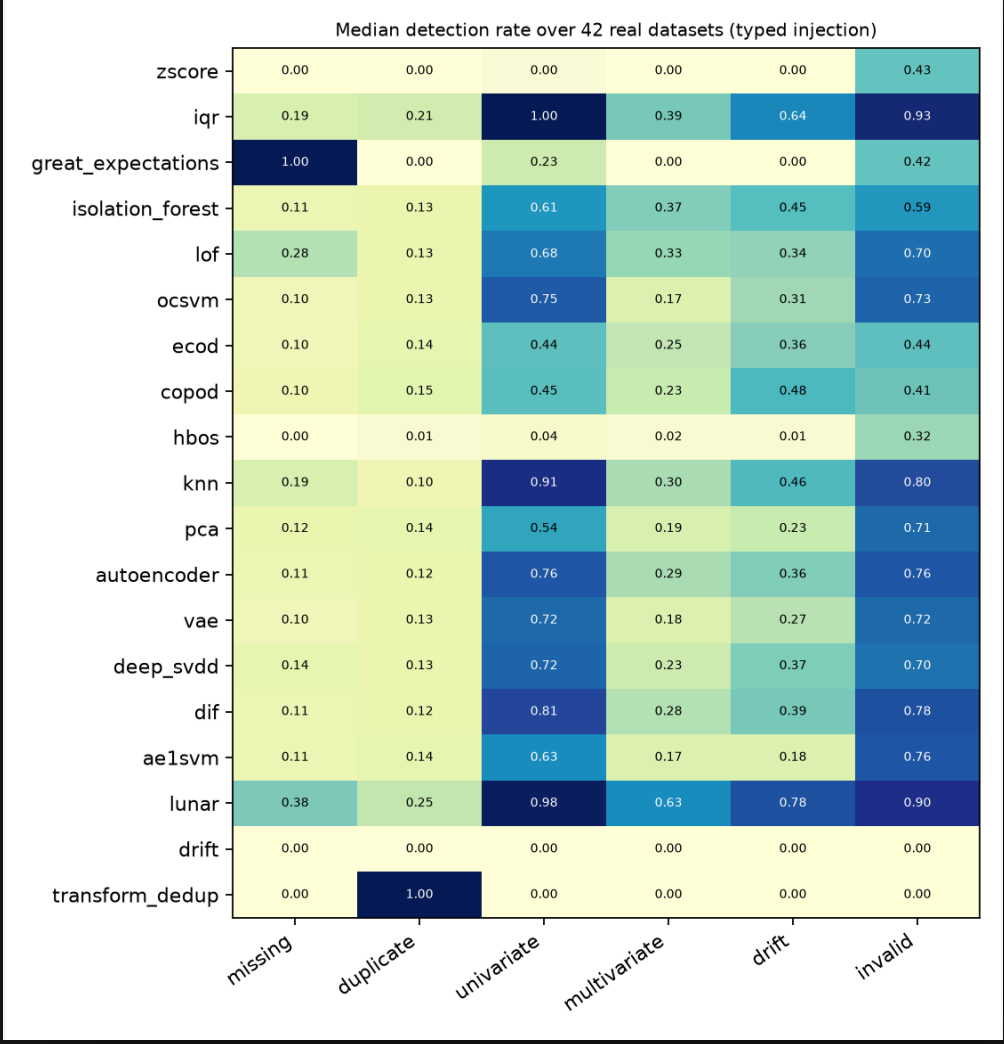
0.61

IQR rule on Annthyroid (best learned model 0.31)

0.96

LUNAR on Mushroom, where the IQR rule is at chance (0.10); label derived by a rule

The synthetic pattern does not simply carry over



Six anomaly types injected into 42 real UCI tables (3% of rows per type, same split and settings as the synthetic run). Protocol and five hypotheses were written first. **Two rows hold everywhere:** Great Expectations finds all missing values, deduplication all duplicates. **Three hypotheses are not supported.**

0.37

Isolation Forest, multivariate (0.89 synthetic)

24%

median clean rows flagged by LUNAR

Median over 42 UCI tables; injected, not natural. LUNAR is best on multivariate (0.63) and drift (0.78).

Two failures that are explained, not hidden

0.00

Z-score, univariate outliers

Median 0.00 against 0.93 synthetic. A check made after seeing it, so exploratory: invalid rows in the training split inflate the standard deviation (**masking**). Without them the Z-score finds every univariate outlier on all 42 tables.

22%

clean rows flagged by IQR

The IQR fences reach the best detection rate on several types, but only by flagging a median 22% of clean rows; the false positive rate reaches 99% on some wide tables. Detection rate has to be read next to it.

0.83

mean rank correlation between two tables (0.70 for drift, partly mechanical)

−0.96

Isolation Forest univariate detection against the number of columns

Injected anomalies, one severity. Deep rows are added on a later slide.

Time changes the answer: a threshold from training does not travel

Detector	PR strat.	PR chrono	ROC chrono	Flag %
Z-score	0.078	0.048	0.935	12.8
IQR	0.018	0.013	0.917	47.1
Great Expectations	0.183	0.045	0.758	3.5
Isolation Forest	0.110	0.035	0.936	0.08

Average precision on the test split; the paper gives a 95% block-bootstrap interval for the chronological arm. The window monitor detects mean shifts of 1 and 2 standard deviations, not 0.5.

Credit Card is the only dataset with a clock. Detectors are fitted on the first 70% of time and scored on the last 15% (52 frauds). Average precision drops for all four detectors, but the interval excludes the stratified mean only for Great Expectations and Isolation Forest.

Prevalence also changes between partitions, so the pilot does not separate it from the effect of time.

0.936

Isolation Forest ROC-AUC, chronological

0.08%

future rows it flags: none of the 52 frauds

`verification/temporal_pilot.md · one two-day dataset: a direction, not a rule`

Part of the gap is only the prevalence

F1 at the threshold	P0 training	P2 rolling
Bank, Z-score	0.51	0.02
Bank, Isolation Forest	0.41	0.11
Credit Card, Z-score	0.00	0.09
Credit Card, Great Expectations	0.00	0.09

Test period, threshold at the training prevalence. Rank correlation of the detector order, stratified against chronological: 0.80 on Bank Marketing, 0.40 on Credit Card. Kyoto-2006+ left out: rights and download not verified.

0.26→0.06

median AP gap on Bank Marketing
after prevalence matching

Credit Card and Bank Marketing (UCI, ordered by date 2008 to 2010), four detectors, three label-free thresholds. On Bank Marketing the prevalence goes from **5.8%** in training to **39.3%** in test: matching the stratified control to it removes most of the gap. A rolling threshold that keeps the training prevalence cannot follow it.

One hypothesis of four holds.

A nominal alert budget is not a promise

Change at a 2% budget	Δ alert / budget [95%]	Δ ROC-AUC
Evaluated scores shifted +1 σ	+4.09 [3.50, 4.77]	0.000
Evaluated scores scaled ×2	+4.86 [4.57, 5.19]	0.000
Noise of 1 σ on the scores	+1.55 [1.32, 1.81]	−0.038
Every feature drifted ½ σ	+20.90 [17.39, 24.55]	−0.051
Monotone change of fit and test scores	+0.001 [0.000, 0.003]	0.000

Budgets of 0.5 to 10% are set from training scores only, on 100 labelled problems. Held-out alert volume was **1.05 to 1.36** times the nominal 1% budget for classical detectors and **1.13 to 2.87** times for neural ones.

+4.1

budgets of added alert volume at 2% (about 5 times nominal) for a 1 σ score shift; ROC-AUC unchanged

Five classical detectors, 98 problems from 78 sources, sources resampled whole, Holm-adjusted. Prevalence reweighted under a label-shift assumption: 0.83 to 1.83 times the budget at 0.5% prevalence, 1.15 to 2.73 times at 20%.

The 30-expected-alert rule for monitoring was chosen after the false alarms were seen. Not promoted to the main tables.

EVERY STUDY ON THE WHOLE POOL

The deeper studies, run on all 152 datasets

	Hypothesis	Result	Value
P1	label-free threshold lowers F1	**holds**	14 of 17 detectors
P2	curve ranking = AP ranking	**fails**	Spearman 0.90 (< 0.99)
P3	recall rises with severity	**fails**	13 of 17; column-wise rules stay at 0
P4	detection share of the pipeline	**holds**	62% (IF), 4% (Z-score), 14 datasets
P5	chronological AP lower	**fails**	31 of 64 pairs; Bank Marketing 0/16
P6	files match the manifest	**holds**	152 of 152

holds fails, reported as failed

Chronology needs a real time variable: 5 labelled datasets qualify (ADBench has none). Each study ran on every dataset where it is defined; the others are listed with a reason, never silently.

58 / 152

tables eligible for typed injection and severity

290 / 304

end-to-end pipeline runs that completed

verification/pool_extension_results.md · six hypotheses written first: three hold, three do not

An LSTM autoencoder is comparable, not clearly better

0.875

LSTM AP on the synthetic stream with broken continuity (row autoencoder 0.006)

0.037

LSTM AP on real Credit Card frauds (row autoencoder 0.024)

Against Isolation Forest (0.035) and Great Expectations (0.045) it is **comparable**. Paired bootstrap, main seed: +0.021 [+0.005, +0.044]; on the other four seeds the interval includes 0.

The window is the ten previous rows of the global stream, since Credit Card has no cardholder. Protocol and a CPU budget of 20 epochs were fixed beforehand.

None of five trainings converged within the 20-epoch budget. Recall of frauds in the later period is at most 0.02. In the per-type matrix the LSTM does not fix drift.

The profile carries over, the values do not

Mean recall, 5 seeds	customers	transactions	sensors
CARRIES OVER EXACTLY			
Great Expectations, missing values	1.00	1.00	1.00
DOES NOT CARRY OVER			
Isolation Forest, multivariate	0.89	0.49	0.32
One-Class SVM, multivariate	0.90	0.18	0.33
best detector, drift	0.50	0.38	0.96

identical on all three depends on the schema

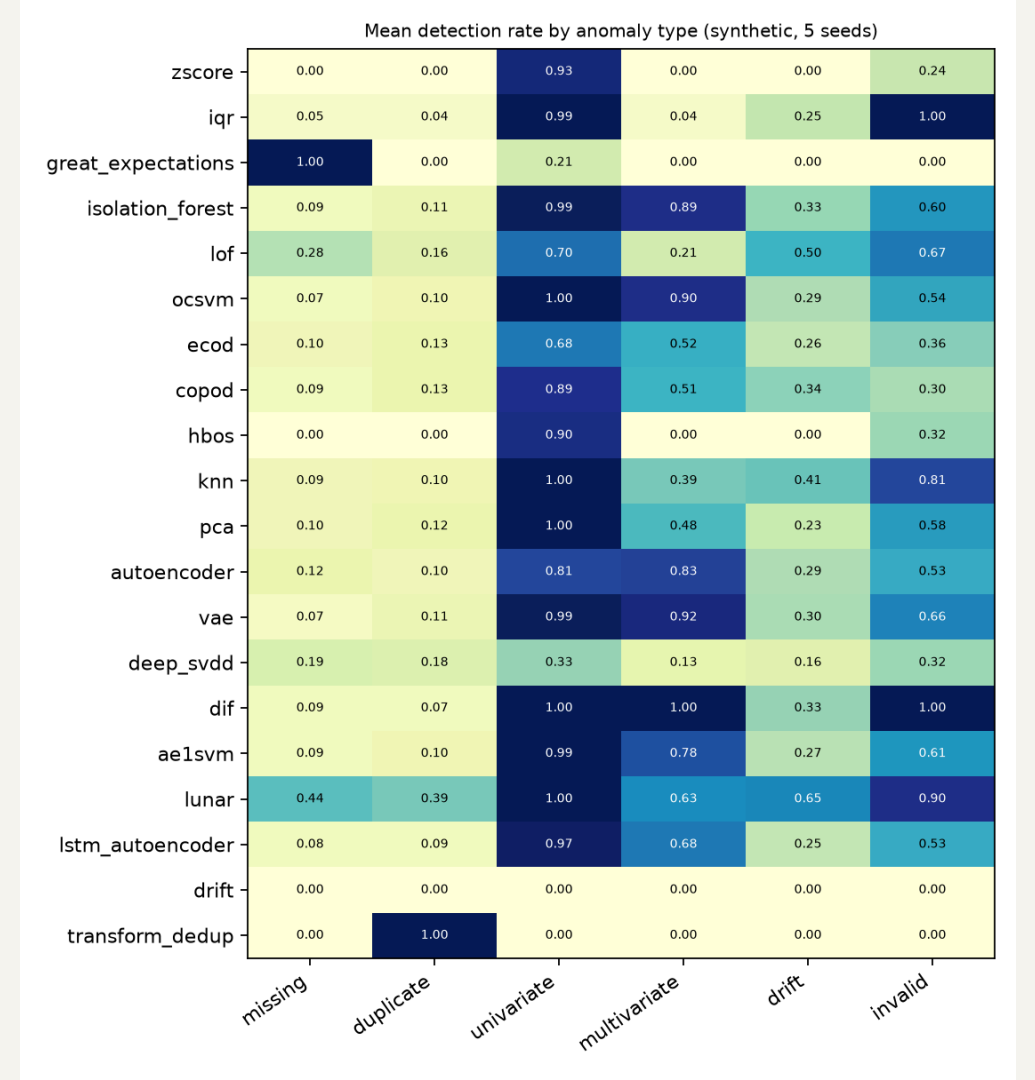
Transactions and sensor readings get the same six types and the same protocol as the customer table. Same generator logic for all three: the agreement shows the profile does not depend on one table, not that it holds on real data. The drift detector flags nothing on any schema.

0.93

mean rank correlation between schemas, first six detectors (0.89 with all seventeen)

Two of four hypotheses hold: the structural rows and the rank stability. Drift and multivariate values depend on how columns are scaled and correlated.

More detectors, same competence map, one exception



Eleven rows added to the synthetic matrix, with three hypotheses written beforehand; **none holds**. ECOD and COPOD find 0.5 of the multivariate outliers although they read one column at a time; PCA reaches 0.48 against 0.83 for the autoencoder.

LUNAR catches 0.65 of the drifted rows (LOF 0.50) while flagging 41% of clean rows. The LSTM still does not fix drift.

0.65

LUNAR drift recall, 41% clean flagged

1.00

DIF recall on multivariate

Synthetic customers are independent rows: drift is not a sequence here.

verification/full_registry.md · research/full_registry.py

Detection latency is not pipeline latency

Detector	10k rows	100k rows	300k rows	Memory MB
Z-score	0.002 s	0.013 s	0.039 s	70
IQR	0.004 s	0.029 s	0.086 s	116
Isolation Forest	0.16 s	1.09 s	3.2 s	140
LOF	0.73 s	18.0 s	**133 s**	405
One-Class SVM	5.6 s	**>300 s**	not run	788
ECOD / COPOD	0.045 s	0.68 s	2.4 s	186
KNN	0.45 s	11 s	68 s	148
VAE	6.8 s	69 s	**209 s**	105
DIF	23 s	**262 s**	**>300 s**	2,748

100 s or more

Median detection-stage time, 8 features; synthetic matrices from 1,000 to 300,000 rows, a fresh process per measurement so peak memory belongs to that run, 300 s budget. Five of the ten newer detectors shown; the paper has all. Single machine, not isolated, not a general ranking.

On the full Credit Card file the Z-score pipeline takes **4.8 s**, of which detection is 3%. Isolation Forest takes **8.5 s**, with detection at 43 to 45%.

133 s

LOF at 300,000 rows (exponent 1.51)

>300 s

One-Class SVM at 100,000 rows

Anomaly type to method

Anomaly type	Use	Recall	Why
missing value	Great Expectations	1.00	not-null expectation, exact
exact duplicate	transform dedup stage	1.00	row equality, not a detector's job
univariate outlier	any statistical rule	0.93 to 0.99	a single-column band is enough
multivariate outlier	Isolation Forest or One-Class SVM	0.89 to 0.90	only the joint models see it (precision about 0.22)
temporal drift	windowed monitor	per-row at most 0.65	sub-threshold per row; LUNAR 0.65 flags 41% of clean rows
invalid format	IQR for the numeric part	1.00 (IQR)	numeric part caught; malformed strings need a format rule

Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. A single aggregate ranking would never surface that.

ACROSS THE THREE EXAMPLE DATASETS

Unsupervised detection works in one of three regimes

Fraud

TOO EXTREME

0.17% positives, and the label mixes every defect. Best F1 of seventeen detectors is 0.35 (Deep SVDD), best precision 0.34 (HBOS).

Cardio

THE FIT

9.6%, rare and distinct. Isolation Forest F1 0.62, AP 0.70. The regime the detectors are built for.

Water

WRONG REGIME

61% majority target. Every ROC-AUC 0.45 to 0.51: the detectors are at chance. A useful negative result.

The question is less "rules or ML" than "does this dataset have rare, outlying anomalies at all". The synthetic dataset sits in the middle at 17.5% and is where the per-type structure is measured. Pipeline cost narrows the shortlist: One-Class SVM and LOF do not survive the full fraud dataset.

Findings from the build

Each of these became a paragraph in the paper. The recurring lesson: the setup around a detector, the operating point, the regime and the pipeline cost matter as much as the algorithm.

The operating point is a lever

Contamination equal to prevalence instead of a fixed 10% moves Isolation Forest fraud precision from 0.01 to 0.20.

One regime out of three

Cardio F1 0.62. Fraud, mixed label: nothing usable. Water Quality: at chance.

ML is weak on missing values

Imputing removes the signal a not-null rule catches for free: GX recall 1.00 against 0.07 to 0.28.

GX 1.x is a full API rewrite

Per-row failures need the COMPLETE result format plus an explicit index column; found by probing the installed library.

PyOD scores can depend on the batch

ECOD, COPOD and DIF use the scored rows for their statistics, the VAE draws a random code, AE1SVM crashes on a last batch of one row. Recorded adaptations, with 17 batch-independence checks; not upstream-equivalence claims.

Drift is not a per-row problem

Once diluted by clean recent rows, the window mean has moved by about a tenth of a standard deviation.

PART 4 OF 4

What the tests protect, and what they do not

Verification of the code that produces the numbers, and mutation testing as a way to see its blind spots.

Illustration, not data: a generated still life.

Historical verification has a defined scope

Check	Result	Scope
line coverage	1,790 / 1,790	pipeline modules and experiment scripts
branch coverage	350 / 350	same scope
mutation, computing code	95.68%	split, metrics, generator, detectors
mutation, input / output	98.87%	readers, writers, scripts
fault injection	23 / 23	four went through every earlier test
real-data check	pass	three CSV files, manuscript tables match

Survivors (122) count as failures and were classified by reading the changes, not by proof.

These measurements belong to local run **d4304c5**. They cover the core pipeline and experiment scripts, with separate verifier checks.

They do not certify the new research modules or establish scientific validity. The run is not independently attested.

1,292
tests, plus 3 on full datasets

97.61%
mutants killed, 4,980 of 5,102

A mutant is a deliberately broken copy of the code

```
# research/alert_budget.py
- flags = scores > info["threshold"]
+ flags = scores >= info["threshold"]
```

This mutant is real. If it were a bug, a score **equal** to the training threshold would raise an alert.

Why bother

Coverage says a line ran. It does not say a test would notice if the line were wrong.

How it works

A tool makes many such copies, one change each, and runs the test suite against every copy.

No test covered this boundary before this session. A new test, "a score equal to the training threshold is not an alert", now fails on the mutant.

The tests are the light. A mutant is killed when a test sees it



A survivor matters because it is either a **behaviour no test protects** (a gap to close with a new test), or a **change that cannot be observed** (an equivalent mutant, next slide). It stays in the count of eligible mutants (the denominator of killed / eligible) until one of the two is shown.

Metaphor, not data: a generated still life. The raised pad casts a shadow; the cracked pad sits flush and casts none.

Some changes cannot be seen, and only the owner may say so

Equivalent mutant

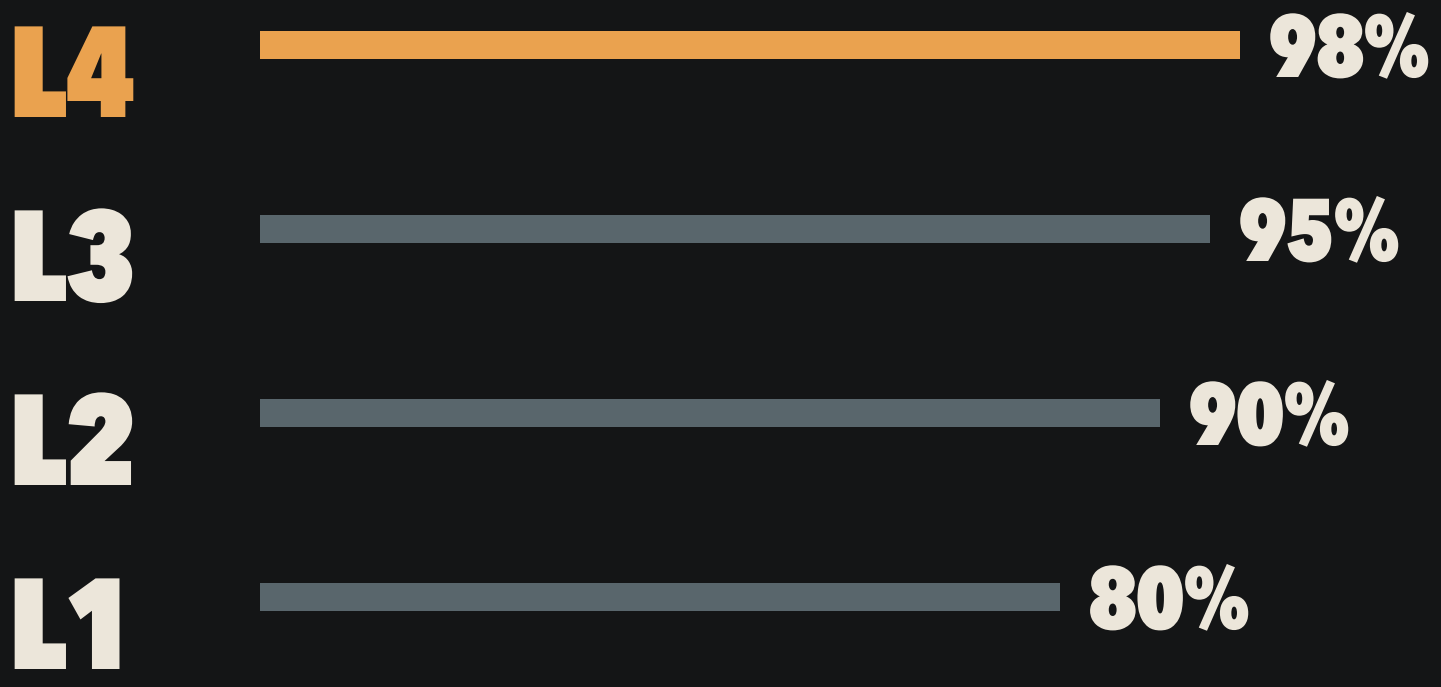
A change that does not alter what the program does, so no test can fail on it. Example: changing `x > 0` to `x >= 0` where x can never be 0. Candidates in this work: a dtype coercion on an array that is already float, `zip(strict=)` on lists of equal length, a `flush=` on a print.

Why approval is required

Calling a survivor equivalent removes it from the denominator and raises the score. If the author of the tests could do that alone, the author would grade their own work. The repository records each equivalence, with its reason, as accepted by the owner.

Status here: the candidates above are **not approved** and stay in the denominator. The four equivalences accepted earlier for the LSTM autoencoder were accepted by the repository owner, who is also the sole author and reviewer: that is owner approval, not independent review.

The bar rises with the scientific risk of the code

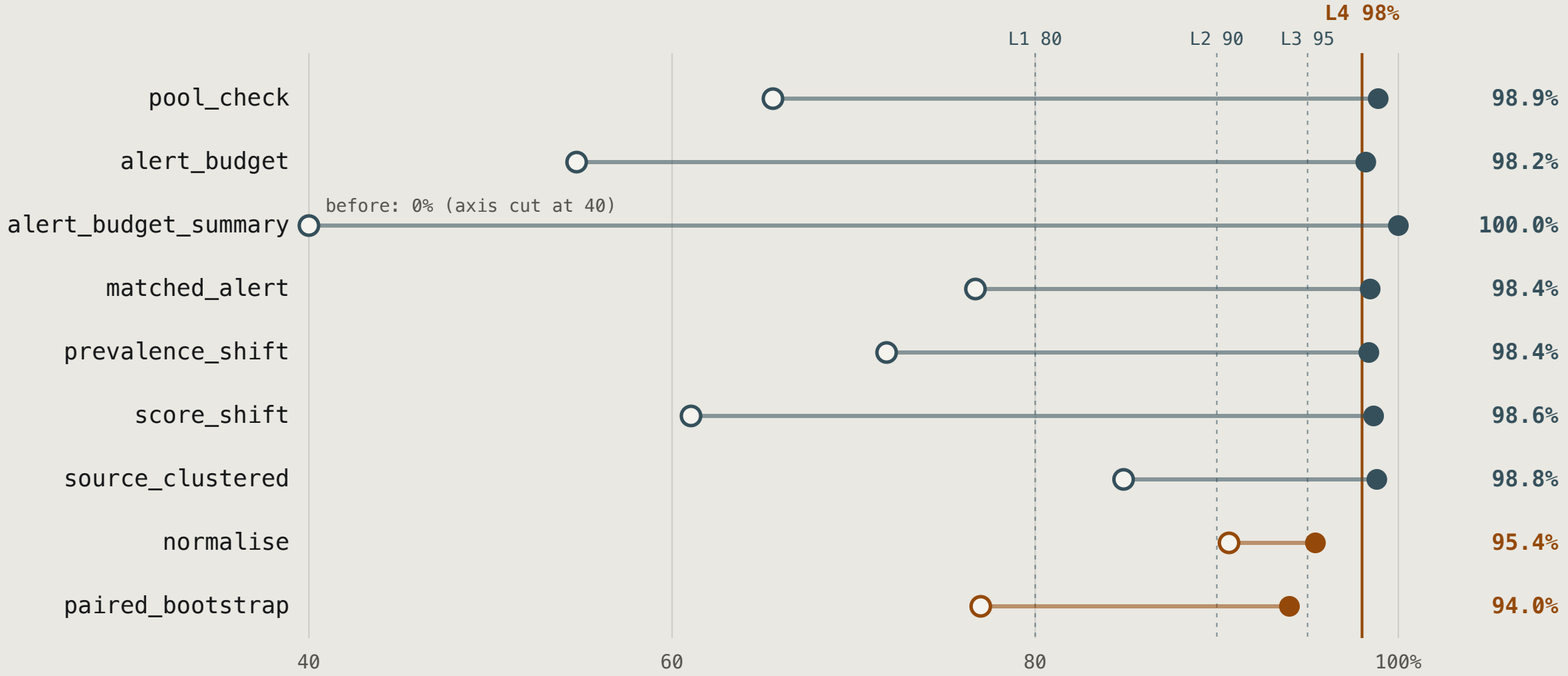


Each module has a risk level, and each level a minimum share of mutants that tests must kill.

L4, the highest, is code whose bug would change a reported result. Its bar is **98%**.

The thresholds were not changed and no module was reclassified.

Nine chosen modules: seven reach the 98% bar, two do not



Local run, 2026-11-11. All nine are L4 modules (9 of 82), chosen by the author. Axis starts at 40%, one module began at 0%. Before: seven from this session's first run, normalise and paired_bootstrap from the 2026-11-07 campaign. Not a certification.

Raw and adjusted scores differ, and the raw one leads

97.61%

raw: 368 of 377 killed

Below the 98% bar for L4. Every survivor counts.

98.66%

adjusted: 368 of 373

Four mutants removed as equivalent. They were accepted by the repository owner, who is the sole author and reviewer: not an independent review.

Five inconclusive mutants stay eligible. No timeout. The adjusted figure depends on a human judgement that raw counts do not need.

WHAT THE EVIDENCE DOES NOT COVER

A focused pass does not certify the repository



26 pass

31 fail

25 incomplete

First complete native campaign, 2026-11-07: 82 L4 modules. 65.5% of L4 mutants killed against 98% required; 7,501 further mutants had no covering test. Bars count modules, the percentage counts mutants, and the 98.66% of the previous slide covers one module of the 82. Not re-measured here.

OPEN

31 L4 modules below threshold and 25 incomplete in that campaign.

OPEN

Survivors that look inert (dtype coercions, zip strict) await an owner decision.

FOUND

The minimum anomaly count, 3, is below the 6 a stratified 70 / 15 / 15 split needs. No archived result changes.

The release gate fails and stays closed. No full campaign was run in this phase.

verification/mutation_risk_policy.json · risk-native-campaign-2026-11-07.json

Experiments run, paper in IMRaD form

DONE

held-out split, 5 seeds, chronological and LSTM pilots, cost study

pool of 152 sets (#66), typed injection on 42 tables (#65)

benchmark on the 100 labelled pool problems

deep rows (#64), multi-schema generator (#67)

IMRaD restructuring (#68, #69, #85); bibliography audit remains #101

decision-level study under shift (#63)

complete registry of 17 detectors, every study rerun

every study extended to the 152-dataset pool, six hypotheses

NEXT: P0 REMEDIATION

Detector and source audits, source-aware statistics, archive replay, publication corrections (issues #89, #91, #100, #103).

The roadmap preserves T001 to T415. **91 of 415** are checked as of 2026-11-11, including T163 to T171 of #94. T172, T062 and T063 remain open, as do the issues.

Optional P2 research is outside the active scope.

TAKEAWAY

Rules and machine learning are complementary by anomaly type.

Across the three examples, unsupervised detection pays off only on Cardio, where anomalies are rare outliers. On 100 labelled pool problems, rankings need train-only revalidation. On 42 real tables with injected types, the synthetic pattern holds only in part, which is why the real-data matrix is the differentiator.

The pipeline, the 152-dataset pool, the protocols and the verification layer are in place. The IMRaD write-up and the remaining studies are what remains.

`github.com/theobigand/etl-anomaly-benchmark · bounded verification · 152 datasets · 17 detectors`